A Summary of Nonlinear Wave Equations

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Contents

1 Overview **2**

I Ideas of Functional and Harmonic Analysis **3**

2 Interpolation Theorem **3**

II Nonlinear Wave Equations **4**

3 Energy Estimates **4**

4 Solutions to the Linear Wave Equation (Constant Coefficients) **4**

5 Solutions to the Linear Wave Equation (Variable Coefficients) **5**

6 Local Theory for Nonlinear Wave Equations **5**

7 Global Regularity for Subcritical Equations **6**

8 More Dispersive Estimates for Linear Wave Equations and the Method of Klainerman **7**

9 Small Data Global Regularity of Solutions **7**

III Incompressible Fluids **8**

10 Introduction to the Euler Equations **8**

11 Construction of Solutions to the Euler Equations **8**

12 Stability of Stationary Solutions of the Euler Equations **9**

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1 Overview

This document is meant to be a very short and concise summary of the main ideas, results, and tecniques developed in the main notes. The emphasis is not on including every material/result studied, but rather the general ideas and intuition (thus representing the "heuristics" behind the theory of nonlinear wave equations and its applications). This document is meant to be read together with the actual set of notes.

Part I
Ideas of Functional and Harmonic Analysis

2 Introductory Harmonic Analysis

In this section, we introduced a lot of new ideas in harmonic analysis that was at the intersection of Fourier analysis and functional analysis. We started off by introducing the notion of *interpolation*. The basic idea behind in

Part II
Nonlinear Wave Equations

3 Energy Estimates

In this section, we studied the linear wave equation $\Box\phi = 0$ on $\mathbb{R}^{n+1}$. In particular, we showed that the linear wave equation has a conserved quantity, the energy $E(t)$, which measures the L^2 size of the solution at the time t. Using the energy function, we can deduce numerous important properties of the linear wave equation. For example, since the wave operator is linear, th difference of two solutions is a solution. Thus since the difference of two solutions has identically vanishing initial data, exploiting energy conservation gave us uniqueness of solutions (even before we showed existence). We can generalize the energy function to linear, *variable*-coefficient wave equations ($\partial_\alpha(a^{\alpha\beta}\partial_\beta) = F$), where instead of conservation, we have a growth bound

$$\sup_{t\epsilon[0,T]} \|\partial\phi\|_{L^2(\mathbb{R}^n)}(t) \lesssim \left(\|(\phi_0,\phi_1)\|_{\dot{H}^1\times L^2} + \int_0^T \|F\|_{L^2}\, dt\right)\exp\left(C\int_0^T \|\partial a\|_{L^\infty}(t)\, dt\right),$$

which tells us that the energy of the ϕ is bounded by the initial energy, the energy introduced by the source term F, and energy introduced by fluctuations in the ambient metric $a^{\alpha\beta}$. This result was proven by expanding out the derivatives, using integration by parts, using the fact that the initial data are all smooth and compactly supported (so that the boundary terms vanish) and concluding using Grönwall's inequality and Young's inequality. Likewise, we established the estimate,

$$\begin{aligned}
&\sup_{t\epsilon[0,T]} \|(\phi,\partial_t\phi)\|_{H^k(\mathbb{R}^n)\times H^{k-1}(\mathbb{R}^n)}(t)\\
&\quad\le C(1+T)\exp\left(C\int_0^T \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t)\, dt\right)\Bigg(\|(\phi_0,\phi_1)\|_{H^k(\mathbb{R}^n)\times H^{k-1}(\mathbb{R}^n)} + \int_0^T \|F\|_{H^{k-1}(\mathbb{R}^n)}(t)\, dt\\
&\qquad + C\int_0^T \sum_{|\alpha|+|\beta|\le k-1} \left\|\partial\partial_x^\alpha a\partial\partial_x^\beta\phi\right\|_{L^2(\mathbb{R}^n)}(t) + \sum_{|\alpha|+|\beta|\le k-1} \left\|\partial_x^\alpha a\partial\partial_x^\beta\phi\right\|_{L^2(\mathbb{R}^n)}(t)\, dt\Bigg).
\end{aligned}$$

4 Solutions to the Linear Wave Equation (Constant Coefficients)

Next, we showed that given compactly supported smooth initial data, there exists a unique solution to the linear homogeneous wave equation with constant coefficients by simply taking the Fourier transform of the wave equation, then solving the resulting ODE. In fact, the weakest function space our initial data needs to be in to guarantee a unique solution is $L^\infty([0,T];H^1(\mathbb{R}^n))\times L^\infty([0,T];L^2(\mathbb{R}^n))$ (the proof of well-posedness just follows from using a mollifier argument). Then, we showed the existence of *forward fundamental solutions*, which are just solutions of the equation $\Box E_+ = \delta$ with zero initial data on the forward light cone. But we can also think about the forward fundamental solution to the linear wave equation with zero source term, and the initial data $(0,\delta)$, where δ is the Dirac delta function at the origin $(x,t) = (0,0)$. We make sense of this heuristically as follows. In the first case, the initial data is zero everywhere, but the $\{t = 0\}$ boundary experiences an infinitesimally short impulse at the origin at $t = 0$. We can think of this being equal to the case where the point at the origin has initial nonzero (even infinite) velocity ∞ at precisely the origin, and zero position everywhere, and zero for all other time $t > 0$. In particular, the fundamental solution is very useful to see that solutions to the wave equation propagate in time, and thus experience dispersion.

Dispersion leads to one of the most important feature of solutions to the wave equation, which is the *decay* of solutions in dimensions $d \ge 2$. The idea is that in dimensions $d \ge 2$, the measure of the support of solutions grows as $t \to \infty$, while the same does not occur in one-dimension. Thus to continue preserving the energy, solutions to the wave equation must decay in time. The proof can follow from using integration by parts, the method of stationary phase approximations, or by the Klainerman-Sobolev inequality.

5 Solutions to the Linear Wave Equation (Variable Coefficients)

In this section, we showed that the linear wave equation with variable coefficients also has a unique solution by first considering the formal adjoint of the wave operator L, then using the Hahn-Banach theorem.

6 Local Theory for Nonlinear Wave Equations

In this section, we looked at the nonlinear wave equation $\partial_\alpha(a^{\alpha\beta}(\phi)\partial_\beta\phi) = F(\phi, \partial\phi)$ with appropriate initial data. We showed that unlike in the linear case where classical solutions exist globally in time, classical solutions to the quasilinear wave equation are guaranteed to exist in the time interval $[0, T_*]$, but not necessarily after the time T_*, called the *maximal time of existence*. However, as with the linear equation, the solution (when it exists) is continuously dependent on the initial data. To prove this result, we wished to exploit the results for linear equations we derived in the previous sections. The Picard iteration scheme is the perfect tool as it essentially linearizes a nonlinear equation. Thus we used this scheme to construct a sequence of smooth functions. Our idea was that if it could be shown that this sequence is uniformly bounded and Cauchy in some Banach space, then the sequence must necessarily be convergent; the bound would then imply that the limit is contained in the correct function space. To uniformly bound the solution, we wanted to use the bootstrap method of assuming that a nice bound on

$$\left\|(\phi^{(i-1)}, \partial_t\phi^{(i-1)})\right\|_{L^\infty([0,T];H^{n+2}(\mathbb{R}^n))\times L^\infty([0,T];H^{n+1}(\mathbb{R}^n))}$$

implies a *worse* bound on

$$\left\|(\phi^{(i)}, \partial_t\phi^{(i)})\right\|_{L^\infty([0,T];H^{n+2}(\mathbb{R}^n))\times L^\infty([0,T];H^{n+1}(\mathbb{R}^n))}.$$

This required us to handle the massive term on the right side of the inequality in the previous page by establishing bounds on the derivatives of F and a, but we were able to get it done. Then to prove Cauchyness of the sequence, we just studied the difference $\phi^{(i)} - \phi^{(i-1)}$, used the mean value theorem, and the conditions on a and F to show that the difference converges to zero. We use a very similar argument to show that the solution is continuously dependent on the initial data.

After this, we showed the persistence of regularity result. Even though our solution may not exist for all time, the persistence of regularity tells us that until whatever maximal time the solution exists, the solution will retain the same regularity as the initial data (i.e., if the initial data is smooth, then the solution will remain smooth up until the maximal time of existence). We were also interested in seeing what happens in the nonlinear case that allows for solutions to potentially blow-up in finite time compared to solutions for the linear equations. The core idea behind this question is Grönwall's inequality. By just considering a simple toy model, one can see that in the linear case, the energy functional satisfies an inequality of the form

$$\frac{d}{dt}E(t) \le f(t)E(t) + g(t).$$

When we use Grönwall's inequality to study this inequality, we compare the energy $E(t)$ to a solution $y(t)$ of the linear ODE $y'(t) = f(t)y(t) + g(t)$. Assuming f, g are nice in special ways, such ODEs always have global-in-time solutions so that $y(t)$, and thus $E(t)$, never blows-up in finite time. Thus it follows that our wave equation has a global-in-time solution. On the other hand, for nonlinear wave equations, the corresponding inequality might be nonlinear in $E(t)$ so that Grönwall's inequality compares $E(t)$ to a solution $y(t)$ of a *non*linear ODE. Unlike with linear ODEs, nonlinear ODEs may have finite-time blow-up. However, in this case, we do not have a certain guarantee that our solution to the wave equation will have finite-time blow up. However, we will not be able to definitively conclude that there is *no* blow-up either. The actual behavior of nonlinear equations is affected by the presence of the derivatives $\partial\phi$, which may not be small or controllable for all times. However, there exist criteria for evaluating whether a solution to a nonlinear wave equation blows up in finite time. This involves looking at the following three terms.

(i)

$$\liminf_{t\to T_*} \|(\phi, \partial_t\phi)\|_{H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n)}(t) \to \infty,$$

(ii)

$$\limsup_{t\to T_*}\left(\sum_{|a|\le\lfloor\frac{n+2}{2}\rfloor}\|\partial_x^\alpha\phi\|_{L^\infty(\mathbb{R}^n)}(t)+\sum_{|\alpha|\le\lfloor\frac{n}{2}\rfloor}\|\partial_t\partial_x^\alpha\phi\|_{L^\infty(\mathbb{R}^n)}(t)\right)\to 0$$

(iii)

$$\limsup_{t\to T_*}\sum_{|\alpha|\le 1}\|\partial^\alpha\phi\|_{L^\infty(\mathbb{R}^n)}(t)\to\infty$$

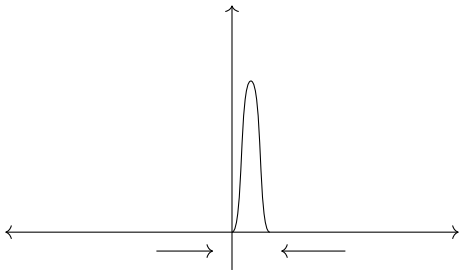
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7 Global Regularity for Subcritical Equations

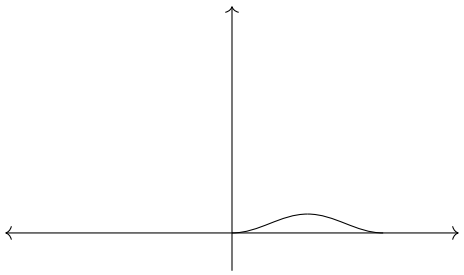
For some nonlinear equations, however, we can say something about global-in-time solutions. In particular, for a toy model, we considered nonlinear wave equations of the form $\Box\phi = |\phi|^{p-1}\phi$ for some $p \ge 2$. In this case, we showed that we have a conserved quantity, namely

$$E(t) = \frac{1}{2}\int_{\mathbb{R}^n}(\partial_t\phi)^2 + (\nabla\cdot\phi)^2 + \frac{1}{p+1}|\phi|^{p+1}\,dx.$$

We also saw that these equations are invariant under certain scaling operations (like $\phi_\lambda = \lambda^{\frac{2}{p-1}}\phi(\lambda t, \lambda x)$). The idea behind looking at scaled solutions is the following: finite-time blow-up solutions look roughly like this:



So if we start with some initially small initial data of the form,



then the scaling above will squeeze the data to look like the wave-form in the top figure. Then we study how the conserved quantity behaves in the limit $\lambda \to \infty$ (i.e., the graph looks more and more like a blow-up solution). If our conserved quantity tends to ∞ (i.e., the conserved quantity gets arbitrarily large), then it is "strong" in the sense that it tries to "fight" against blow-up. This is the "definition" of subcriticality. In some cases, the conserved quantity might scale with λ. In this case, the conservation law does not give us enough information (i.e., we cannot conclude that there is no blow-up, but we also cannot conclude that there is a blow-up); this is the critical case. Finally, in the supercritical case, the conserved quantity actually becomes smaller in the limit. Thus we have even less information about what is happening. In particular, one has that given the right initial data for the PDE $\Box\phi = |\phi|^2\phi$, the solution is global-in-time (one can show that this equation is subcritical).

8 More Dispersive Estimates for Linear Wave Equations and the Method of Klainerman

In this section, we returned back to a discussion of the linear wave equation $\Box\phi = 0$ on $\mathbb{R}^{n+1}$ $(n \geq 2)$ with some appropriate initial data. Previously, we showed that solutions to this equation experienced a uniform sharp decay rate of $\mathcal{O}(t^{\frac{1-n}{2}})$, and on bounded intervals a better decay rate of $\mathcal{O}(t^{-n/2})$. We proved these estimates using just integration by parts. But we haven't yet used the power of the energy to establish the bounds. Thus this section focused on introducing the energy methods of Klainerman that enables these types of analysis. At the heart of Klainerman's approach is the realization that the wave equation is invariant under certain transformations, like spatial and temporal translations, spatial rotations, and Lorentz boosts. Intuitively, this is because Minkowski space is homogeneous. The running notation convention is that Γ will always depict one of the vector fields under which the wave equation is invariant. All of the energy estimates that we have proven for the linear wave equation continue to hold for these solutions $\Gamma^\alpha\phi$ of $\Box\phi = 0$. Now we recall the standard Sobolev embedding which states that if $s > n/2$, then $L^\infty \hookrightarrow H^s$ continuously. In particular, for $n \geq 2$ (so that $n/2 \geq 1$), if $\phi \in H^{\lfloor \frac{n+2}{2} \rfloor}$, then the Sobolev embedding theorem tells us that $\|\phi\|_{L^\infty} \lesssim \|\phi\|_{H^{\lfloor \frac{n+2}{2} \rfloor}}$; i.e., ϕ is bounded. But in general, we are interested in learning more about ϕ (e.g., does it disperse or decay?) Thus the idea of Klainerman-Sobolev was that if instead of just considering the standard Sobolev embeddings, by considering the symmetries of $\Box$, one can look at a "weighted" Sobolev space argument, then obtain key dispersion/decay estimates for the solution. This is encapsulated by the Kleinerman-Sobolev inequality, which states that

$$\|\phi(t,\cdot)\|_{L^\infty(\mathbb{R}^n)} \lesssim \frac{1}{(1+t+|x|)^{\frac{n+1}{2}}(1+|t-|x||)^{1/2}} \cdot \sum_{|\alpha| \leq \lfloor \frac{n+2}{2} \rfloor} \|\Gamma^\alpha\phi\|_{L^2(\mathbb{R}^n)}.$$

In particular, we are able to convert L^2 information about the derivatives of ϕ into a pointwise estimate on the solution. Moreover, unlike the standard Sobolev embedding, this inequality also gives us information about the *decay* rate of the solution.

9 Small Data Global Regularity of Solutions

While in general supercritical equations are hard to create a general theory (or even a generic statement about what solutions behave like) since so less is known about these equations, under certain specific conditions we can say a lot of important ideas. In this section, we looked at the *wave map equation*,

$$\Box\phi = \phi\left((\partial_t\phi)^\intercal(\partial_t\phi) - \sum_1^4 (\partial_i\phi)^\intercal(\partial_i\phi)\right).$$

The idea is that the "bad" behavior in nonlinear equations is caused by the nonlinearities due to quadratic terms consisting of the derivatives of ϕ growing too "large". Thus the idea is that if we can control the growth of these nonlinear terms, then the wave map equation will essentially act like a linear PDE, for which we have global-in-time smooth solutions. Thus we made certain assumptions on how "large" the derivatives of ϕ are. In particular, we assumed that the initial data are smooth and compactly supported inside balls of radius $R > 0$. We also ensured that the derivatives of the initial data weren't too large; i.e., that

$$\sum_{|\alpha| \leq k} \|\partial\partial^\alpha\phi_0\|_{L^2(\mathbb{R}^4)} + \|\partial^\alpha\phi_1\|_{L^2(\mathbb{R}^4)} < \varepsilon,$$

where $k \geq 6$ and $\varepsilon > 0$. Then what we found was that the solution exists for *all* time, *and is smooth for all time* provided that ε was sufficiently small enough. The proof used the following ideas: (1) we show that some type of weighted energy is bounded above by a ("bad") constant, (2) exploit the Klainerman-Sobolev inequality to show that this bound can be improved, (3) use continuity to show that the solution exists for all time and is smooth (i.e., we used a bootstrapping argument). However, a key part of the argument was the integrability of $1/(1+t)^{3/2}$ in time. An analogous proof in $\mathbb{R}^{3+1}$ would yield a term with $1/(1+t)$, which is not integrable. Thus a different technique needs to be used.

Part III
Incompressible Fluids

10 Introduction to the Euler Equations

In this section, we saw our first introduction to the *incompressible Euler equations*, which are a system of quasilinear hyperbolic PDE for the velocity field u of an *ideal* fluid moving in some domain $\Omega \subseteq \mathbb{R}^n$, where we will almost always assume $n = 2, 3, 4$. These equations were made under a few important hypotheses. First, we assumed that the motion is **incompressible**, which is to say that the density of a fluid particle along a flow line of the velocity field remains constant in time (i.e., $D_t\rho = 0$, where $D_t = \partial_t + (u \cdot \nabla)$ denotes the **material/convective derivative**). Thus this adds an important constraint, namely that the velocity field of the fluid is divergence free (i.e., $\nabla \cdot u = 0$). For the intents and purposes of this book, we assumed that the density of the fluid was a constant for all spacetime points (x, t). Another important hypothesis we operated under was the fact that the fluid is **ideal**; i.e., the only interactions between two distinct fluid particles is any interaction arising out of the incompressibility clause (in particular, this implies that the **visocity** of the fluid is zero). Thus, the Euler equations for an incompressible fluid can be written as the following,

$$\begin{aligned}
&\partial_t u + (u \cdot \nabla)u = 0,\\
&\nabla \cdot u = 0, (u \cdot \hat{n})|_{\partial\Omega} = 0,\\
&u|_{\{t=0\}} = u_0.
\end{aligned}$$

One can derive these equations explicitly from a Lagrangian (i.e. calculus of variations) principle.

Then we looked at some basic properties of the Euler equation. One of the features we were interested in was learning how fluid motion works, at least on a local scale. To that end, we defined the **vorticity** ω of the fluid to be the curl of the velocity field, $\omega = \nabla \times u$, which roughly measures how much the fluid rotates. Then by taking the Taylor expansion of the velocity field locally, we observe that the field u can be broken up into three component motions: (1) a linear translation, (2) a deformation (i.e., stretching and shrinking), and (3) a rotation (via the vorticity). Under specific hypothesis on the domain Ω, it then follows that knowing the vorticity field of a fluid ω enables us to deduce the velocity field u via the **Biot-Savart Law**. Indeed, for $\mathbb{R}^2$ and simply connected domains $\Omega \subset \mathbb{R}^2$, Poincaré's theorem tells us that if u is divergence-free (as is the case for incompressible fluids), then there exists a function ψ such that $u = \nabla^\perp \psi$. Then taking the Laplacian of ψ, we obtain $\Delta\psi = \omega$, which is precisely a Poisson equation (i.e., a second-order elliptic equation). Thus by the theory of elliptic condition, if our domain Ω satisfies certain properties, then ω can be uniquely (up to a constant) determined, whence u can also be determined via the **Biot-Savart law**, $u = \nabla^\perp \Delta^{-1}\omega$. In particular, we showed this was possible in the case of a simply connected bounded domain of $\mathbb{R}^2$ and for $\mathbb{R}^2$ itself. Our next order of business was to establish some conservation laws, of which the conservation of energy $E(t)$, where

$$E(t) \coloneqq \frac{1}{2}\int_\Omega u^2(t)\, dx,$$

and the conservation of the vorticity along stream lines (in $\mathbb{R}^2$) was conserved. The former conservation law becomes essential when trying to establish regularity theorems for solutions of the Euler equation, and the latter when trying to understand differences in the results of existence theorems to the Euler equations.

11 Construction of Solutions to the Euler Equations

While in the previous section, we introduced the Euler equations and saw some of their properties and hinted at ways we could prove existence and uniqueness (e.g., by showing that solving the equations uniquely was equivalent to solving for the vorticity uniquely), in this section, we explicitly carried out some of these computations. We started by trying to figure out a *global* existence and uniqueness theorem for the Euler

equations in $\Omega = \mathbb{R}^2$. As we showed in the previous section, knowing the asymptotic behavior of the velocity field u and the vorticity field ω is sufficient to deduce the existence and uniqueness of the velocity solving the Euler equations. So we looked at the Euler equations in the vorticity formulation,

$$\begin{aligned}
&\partial_t + (u \cdot \nabla)\omega = 0, \\
&\omega = \nabla \times u, \quad \nabla \cdot u = 0, \\
&u \cdot n = 0, \quad u|_{\{t=0\}} = u_0.
\end{aligned}$$

Then to prove global existence and uniqueness for the Euler equations, it sufficed to demonstrate existence and uniqueness of the triple (Φ_t, u_t, ω_t) where Φ_t solves the ODE $\frac{d}{dt}\Phi_t = u(\Phi_t, t)$ with $\Phi_0(x) = x$, $\omega_t = \omega_0(\Phi_{-t}(x))$, and u_t is given by the convolution $K_\Omega * \omega_t$, where K_Ω is the Green's function for Ω. As we did for the general nonlinear wave equation above, we used a Picard iteration scheme to obtain a sequence of approximate solutions. Then we used some of the regularity estimates for Sobolev and L^p functions to show that the sequences for Φ_t, u_t, and ω_t we constructed actually do converge in the appropriate function spaces. Thus this gave us *weak solutions* to the Euler equations. But since our equations are of physical origin, we desire *classical solutions*. We then presented an analogous "persistence of regularity theorem" for the Euler equations, which stated that as long as our initial datum for the vorticity had some regularity, then our solution for ω was a classical solution that remained in the same vorticity class. Then we wanted to turn to the Euler equations in 3D. Unlike in the two-dimensional case, we quickly saw that our unique solution does not necessarily exist globally-in-time, and could have finite-time blowup. We showed this heuristically by considering the vorticity. In two dimensions, conservation of the vorticity along stream lines drastically simplified the equations we were looking at. On the other hand, in the three-dimensional case, the Jacobian $\nabla\Phi_t$ acts on the vorticity ω_t along each stream line via the term $(\omega \cdot \nabla)u$. But in some sense, ω is like the antisymmetric part of ∇u. Thus $(\omega \cdot \nabla)u \sim \omega^2$. This means that the Euler equations in the vorticity formulation resembles the ODE $\dot{y}(t) = y^2$, with initial data that is positive everywhere. This ODE has finite-time blowup. Thus by comparison, it follows that solutions to the Euler equations *could* also develop blow-ups in finite time. Therefore, the proof focused on constructing a *local-in-time* unique solution. As with the two-dimensional case, we began the construction by constructing a sequence of approximate solutions by means of projecting onto the subspace of $L^2(\Omega)$ generated by functions of the form $e^{ik\cdot x}$. Then after taking the Fourier transform of the Euler equations in the spatial coordinate, we obtained an ODE in time. The idea behind this is the *Faedo-Galerkin method*[1] which states that to prove existence to a certain nonlinear PDE, it suffices to show that the solution arises as the limit of approximate solutions to an orthogonal and complete system in the desired L^p/regularity class. Thus since the eigenvalues of the Laplace operator form a complete orthogonal system in $L^2(\Omega)$, we choose to project the nonlinear PDE down into finite-dimensional subspaces generated by finitely many of these eigenfunctions. In doing so, we reduce the system down to a coupled system of nonlinear ODEs. Standard ODE theory then gives us an approximate solution for each N. The *a priori* energy estimates for the Euler equations enables us then to show that these approximate solutions are bounded in the desired function spaces, thus also allowing us to conclude that the constructed sequence converges (uniformly on compacta). Then a standard argument shows that the limit is not only a weak solution to the Euler equations, but also a classical solution.

12 Stability of Stationary Solutions of the Euler Equations

[1] https://math.stackexchange.com/questions/4846360/faedo-galerkin-to-2d-pde